

CAPE POINT, South Africa

WMO: 689160

Lat: 34.3533S

Lon: 18.4894E

Elev: 228

StdP: 98.62

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	9.3	0.0	3.9	17.9	2.0	4.5	15.4	21.4	12.9	19.9	12.1	7.3	230	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.2	27.0	17.5	25.0	17.5	23.5	17.5	20.4	23.2	19.7	22.4	19.1	21.7	5.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.7	21.3	18.8	14.0	20.8	18.2	13.5	20.4	59.6	23.6	57.4	22.4	55.4	21.8	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.9	18.5	7.1	32.0	0.9	2.4	6.4	33.7	5.9	35.1	5.4	36.5	4.7	38.2		
(5)			5.0	21.7	0.9	0.7	4.4	22.2	3.9	22.7	3.4	23.1	2.8	23.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	19.3	19.4	18.4	17.0	15.4	13.8	13.2	13.1	13.9	15.3	16.4	18.4
	DBStd	3.02	1.52	1.58	2.04	2.55	2.02	1.98	2.26	2.01	2.18	2.02	1.77	1.80
	HDD10.0	3	0	0	0	0	0	1	1	1	0	0	0	0
	HDD18.3	959	6	6	22	56	94	139	159	164	136	98	61	20
	CDD10.0	2241	289	264	262	209	168	113	102	96	118	164	193	262
	CDD18.3	156	37	36	25	16	4	2	1	1	3	4	4	23
	CDH23.3	387	51	53	91	82	14	4	2	3	26	20	12	30
	CDH26.7	83	10	7	28	21	1	0	0	0	7	2	2	4
Wind	WSAvg	7.8	7.5	8.7	8.5	8.4	7.5	7.7	7.6	7.6	7.0	7.7	7.8	7.6
Precipitation	PrecAvg	670	21	16	22	52	90	101	111	109	63	34	33	20
	PrecMax	893	96	35	73	129	167	182	240	204	150	126	99	42
	PrecMin	442	0	0	0	12	6	27	40	51	21	1	2	0
	PrecStd	123	22	10	18	28	36	33	48	46	32	25	26	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	27.7	30.2	29.3	25.6	23.4	22.9	23.0	26.9	26.2	24.7	26.5
		MCWB	18.7	18.8	19.0	16.8	15.7	13.7	12.6	13.4	14.6	16.3	16.9	19.4
	2%	DB	24.5	24.7	25.8	25.4	22.7	20.7	20.6	19.9	21.2	22.5	21.7	23.9
		MCWB	19.4	18.7	18.1	16.1	14.0	12.1	12.1	11.6	13.3	15.6	16.7	19.0
	5%	DB	22.9	23.2	22.9	22.7	20.5	18.4	18.6	17.3	18.1	19.7	20.2	22.5
		MCWB	18.8	18.6	18.1	15.9	13.7	11.5	11.6	12.2	13.6	15.3	16.3	18.2
	10%	DB	21.9	22.0	21.3	20.5	18.5	16.5	16.4	15.7	16.5	18.2	19.1	21.3
		MCWB	18.3	18.2	17.5	16.0	13.9	12.5	12.1	12.4	13.2	14.9	15.8	17.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	20.7	20.6	19.0	17.2	15.5	15.0	15.1	16.5	18.1	18.8	20.8
		MCDB	23.9	23.6	24.8	23.6	21.0	18.5	17.3	17.8	21.8	22.6	22.3	24.4
	2%	WB	20.5	20.0	19.6	18.0	16.2	14.6	14.1	14.2	15.2	16.8	17.6	19.7
		MCDB	22.7	22.5	23.0	21.7	18.7	16.8	16.3	16.4	18.5	20.0	20.4	22.8
	5%	WB	19.7	19.4	18.8	17.3	15.5	14.0	13.4	13.6	14.4	16.0	16.9	19.0
		MCDB	21.9	21.9	21.6	20.4	17.7	15.8	15.6	15.6	17.1	18.7	19.4	21.6
	10%	WB	19.0	18.8	18.2	16.6	14.9	13.5	12.9	13.0	13.7	15.3	16.2	18.3
		MCDB	21.1	21.3	20.8	19.5	17.0	15.3	15.0	14.9	16.0	17.7	18.5	20.6
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.5	5.4	5.4	4.6	4.2	4.3	4.5	4.7	5.0	5.1	5.3
		MCDBR	6.7	7.6	8.3	9.5	8.2	7.7	7.9	7.6	8.0	7.9	6.8	6.9
		MCWBR	3.4	3.6	3.7	3.9	3.4	3.8	3.6	3.9	3.8	3.8	3.8	3.6
	5% WB	MCDBR	5.8	6.4	6.7	7.1	5.5	4.8	5.1	5.4	6.5	6.8	6.3	6.4
		MCWBR	3.3	3.6	3.6	3.7	3.4	3.4	3.4	3.5	3.7	3.7	3.9	3.6
Clear-Sky Solar Irradiance	taub	0.336	0.338	0.328	0.317	0.316	0.310	0.312	0.325	0.335	0.344	0.333	0.337	
	taud	2.580	2.581	2.606	2.630	2.618	2.605	2.599	2.562	2.498	2.488	2.564	2.566	
	Ebn at Noon	1001	978	952	909	853	830	848	885	929	960	998	1003	
	Edh at Noon	105	101	92	80	72	68	72	84	101	110	106	107	
All-Sky Solar Radiation	RadAvg	8.06	7.16	5.69	4.17	2.89	2.42	2.69	3.46	4.79	6.33	7.53	8.18	
	RadStd	0.17	0.22	0.24	0.20	0.17	0.14	0.15	0.17	0.20	0.28	0.24	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.36	+0.49	N/A	N/A	+0.53	+0.47	-2	-74	+111	+61

Nomenclature: See separate page